6. Blood Donor & Camp Management – Automation Suite & Workflow Options

Aim

To design and document the automation layer for the Blood Donor & Camp Management application using Zoho Creator workflow options, focusing on form validation, donation records, blood inventory, and blood request processing.

Objective

This experiment demonstrates how Zoho Creator workflows can improve the management of donor registrations, blood donation camps, donation records, blood inventory, and blood requests. The experiment includes form validation and explores how additional workflow options can be used to automate routine operations.

Workflow Mapping Overview

Workflow Type	**Configuration / Purpose**
Form Validation	Validate_Blood_Request – checks the units required and units issued before accepting a blood request.
Form Workflow	Update_Inventory_After_Donation – proposed workflow to update inventory after a successful donation; kept disabled to prevent incorrect or duplicate updates.
Form Workflow	Update_Inventory_After_Blood_Issue – proposed workflow to deduct issued units from inventory; kept disabled until safe inventory-update logic is implemented and tested.
Schedule	Pending_Blood_Request_Reminder – proposed schedule for identifying pending blood requests and sending reminders.
Approval	Blood_Request_Approval – proposed workflow for reviewing and approving blood requests.
Blueprint	Blood_Request_Lifecycle – proposed process for controlling request stages from submission to completion.
Function	Blood_Request_Processing_Function – proposed reusable function for request-processing operations.

Detailed Workflow Configurations & Evidence

Form Validation – Validate_Blood_Request

- **Purpose:** Ensures that blood request records contain valid unit quantities before submission.
- **Configuration:** Created under Blood Requests using the form submission validation workflow.
- **Validation rules:**
  - Units Required must be greater than zero.
  - Units Issued cannot be negative.
  - Units Issued cannot exceed Units Required.
- **Action:** Displays a validation message and prevents submission when the entered values violate these rules.
- **Testing:** A request with 3 units required and 4 units issued was tested and correctly rejected by the validation workflow.

Form Workflow – Update_Inventory_After_Donation

- **Purpose:** Intended to increase the available blood inventory after a donation is recorded as collected.
- **Proposed action:** Identify the donor's blood group, locate the matching inventory record, add the collected quantity, and update the last-updated date.
- **Current status:** The workflow was kept disabled because inventory updates must be tested carefully to prevent duplicate counting.

Form Workflow – Update_Inventory_After_Blood_Issue

- **Purpose:** Intended to deduct issued blood units from the corresponding inventory record when a request is partially or fully completed.
- **Proposed action:** Identify the requested blood group, calculate the units issued, update the available inventory, and record the update date.
- **Current status:** The workflow was kept disabled until duplicate deductions and insufficient-stock conditions can be handled safely.

Schedule – Pending_Blood_Request_Reminder

- **Purpose:** Proposed automation for identifying blood requests that remain pending.
- **Proposed action:** Run at a configured interval to identify pending requests and notify the responsible administrator where notification functionality is configured.
- **Current status:** Proposed; not yet implemented.

Approval – Blood_Request_Approval

- **Purpose:** Proposed approval process for reviewing blood requests before they are fulfilled.
- **Proposed stages:** Request submitted, under review, approved or rejected, and completed.
- **Current status:** Proposed; not yet implemented.

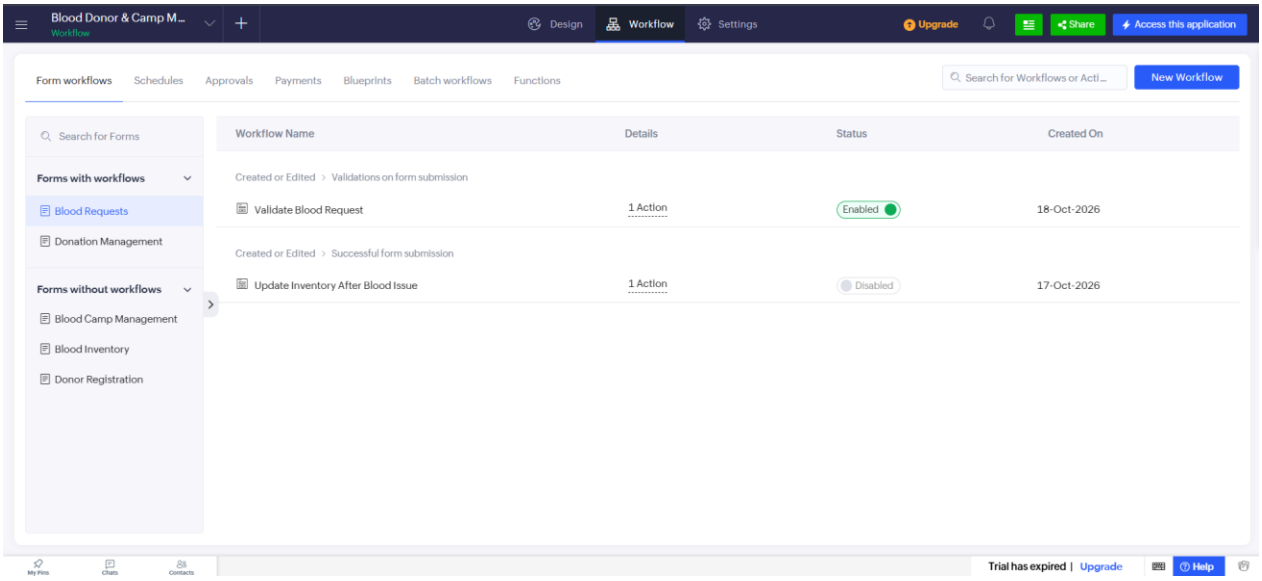
Blueprint – Blood_Request_Lifecycle

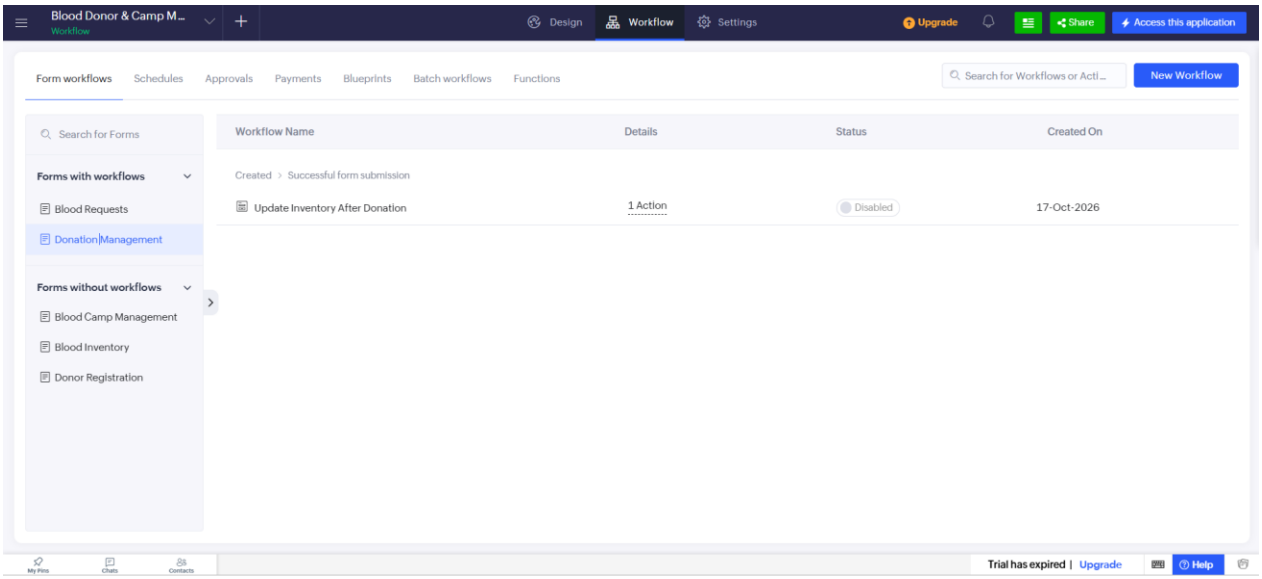
- **Purpose:** Proposed use of a Blueprint to control the progression of blood requests through defined stages.

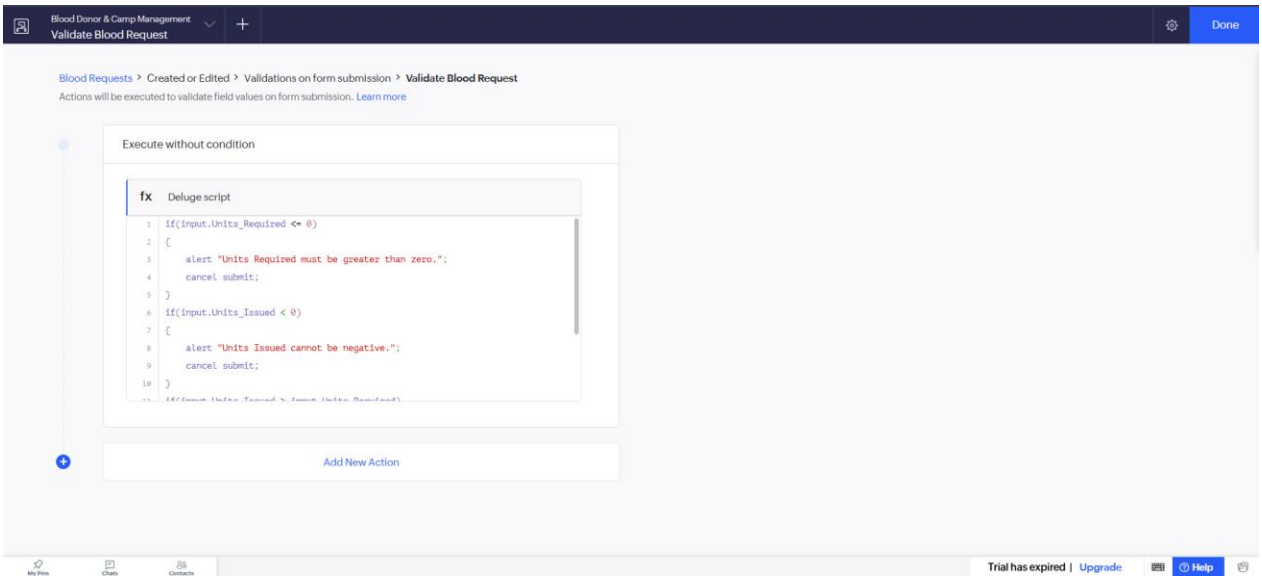
- **Proposed transitions:** Pending → Under Review → Approved or Rejected → Partially Issued or Completed, depending on the request outcome.
- **Current status:** Proposed; not yet implemented.

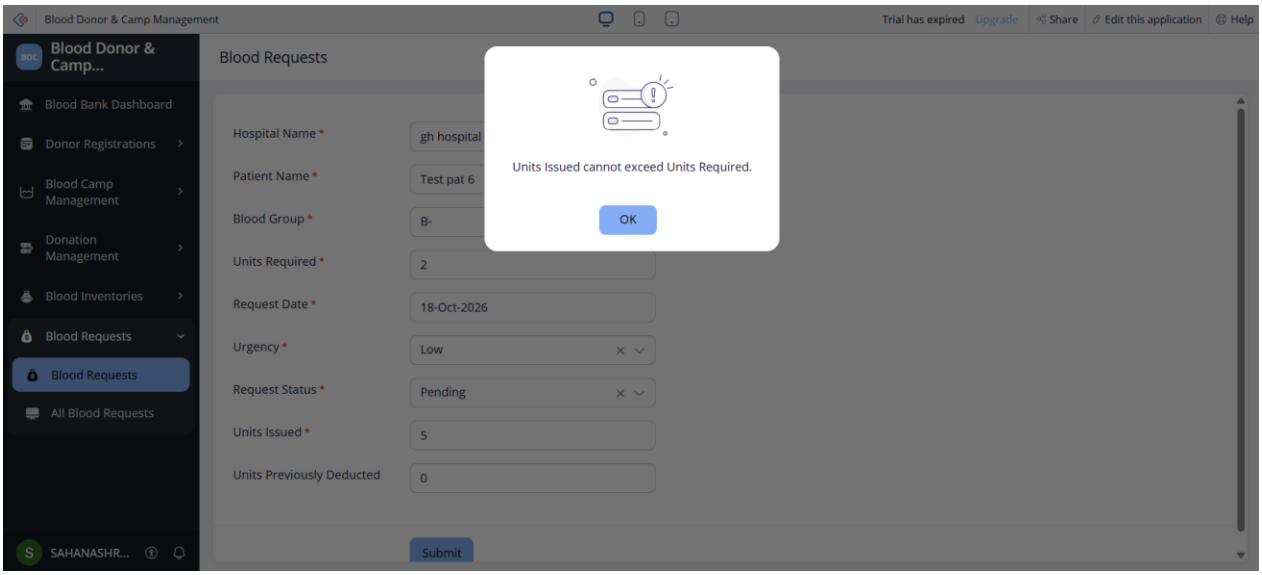
Function – Blood_Request_Processing_Function

- **Purpose:** Proposed reusable Deluge function for handling repeated blood request processing tasks.
- **Proposed operations:** Validate request quantities, check the request status, and support inventory processing after suitable safeguards are implemented.
- **Current status:** Proposed; not yet implemented.









Result

The Blood Donor & Camp Management application was configured with a working blood request validation workflow in Zoho Creator. Testing confirmed that invalid unit quantities were rejected during submission. Additional automation options for inventory updates, reminders, approvals, request lifecycle management, and reusable functions were identified for future implementation.